Assignment 2 Report — CO2

Data Mining Preprocessing, Dispersion, Cleaning, Integration, Transformation, Reduction & Validation

1. Introduction and Objectives

Data preprocessing is widely recognized as the most time-consuming yet vital phase of data mining. Raw datasets harvested from operational systems and IoT sensors are rarely ready for algorithmic analysis; they are burdened by missing values, noise, outliers, and schema inconsistencies. In this assignment report, our team has documented thorough methodological solutions, manual calculations, algorithmic mappings, and experimental validations across seven distinct data mining scenarios.

2. Matching Business Problems to Mining Functionalities

For TeleNova Telecom (managing 8 million active subscribers), we mapped six operational challenges to appropriate data mining functionalities, learning paradigms, and candidate algorithms:

Problem	Business Scenario	Mining Functionality	Learning Type	Candidate Algorithms
P1	Which telecom add-ons are purchased together?	Association Rule Mining	Unsupervised	Apriori, FP-Growth
P2	Which prepaid subscribers are likely to churn?	Classification	Supervised	Decision Tree (J48), SVM
P3	What are the natural subscriber usage clusters?	Clustering	Unsupervised	K-Means, DBSCAN
P4	Which accounts exhibit abnormal SIM card usage?	Outlier Detection	Unsupervised	Isolation Forest, LOF
P5	What characterizes high-value vs low-value users?	Concept Description	Supervised	AOI, Random Forest
P6	How does monthly data usage trend over 24 months?	Evolution Analysis	Unsupervised	ARIMA, PrefixSpan

Critical Evaluation: The marketing manager's suggestion to purchase a single "all-purpose mining algorithm" was respectfully rejected. Association, classification, clustering, outlier detection, and evolution analysis rely on fundamentally distinct mathematical principles and cannot be solved by one universal algorithm.

3. The KDD Process, Architecture and Mining Challenges

MediCore Hospital Readmission Scenario: We mapped the Knowledge Discovery in Databases (KDD) process through its five iterative stages (Selection, Preprocessing, Transformation, Data Mining, Evaluation). We designed a modular data mining architecture and addressed five key challenges: handling severe class imbalance (only 6% readmissions), ensuring user interpretability for clinical staff, scaling to millions of patient records, processing multi-modal electronic health records, and safeguarding patient privacy.

Rebuttal: To the hospital director who argued that SQL queries and standard statistical reports are sufficient, we clarified that SQL provides retrospective descriptive summaries ("what happened"), whereas data mining uncovers predictive risk patterns ("what is likely to happen and why").

The Knowledge Discovery in Databases (KDD) Process



4. Describing and Comparing Data Dispersion

Delivery Time Dataset (24 observations in minutes): 28, 31, 33, 35, 35, 36, 38, 40, 41, 42, 42, 43, 45, 46, 48, 50, 52, 53, 55, 58, 61, 64, 72, 95.

• **Central Tendency:** Mean = 47.38 mins | Median = 44.0 mins | Mode = 35 & 42 mins | Midrange = 61.5 mins.

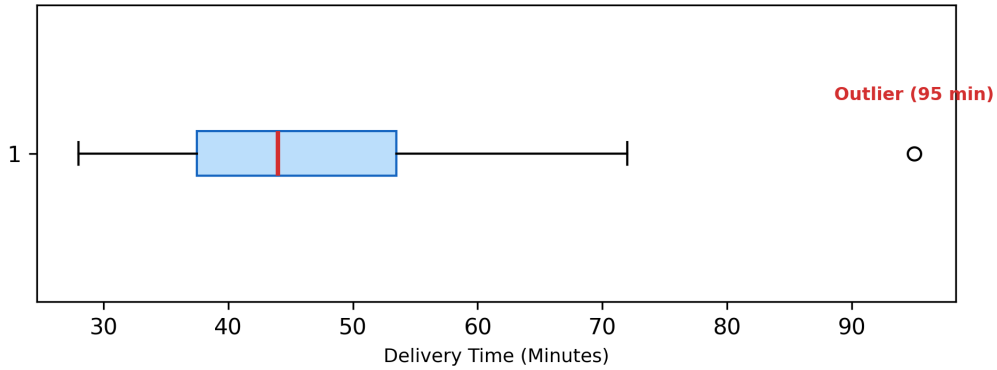
• **Spread & Quartiles:** Q1 = 36.5 mins | Q3 = 54.5 mins | IQR = 18.0 mins.

• **Variance & Std Dev:** Variance = 323.97 min^2 | Standard Deviation = 18.0 mins.

• **Outlier Detection:** Upper fence Q3 + 1.5(IQR) = 81.5 mins. The 95-minute delivery time is an extreme outlier.

• **Recommendation:** Because the distribution is right-skewed by extreme delivery delays, operations should report the **Median (44.0 mins)** as the typical delivery time and **IQR (18.0 mins)** for variability.

E-Commerce Delivery Times Dispersion & Outlier Detection



5. Cleaning Smart-Meter Data & Smoothing

We examined smart-meter energy readings (M101–M106), identifying missing values (M102, M105), text formatting errors (M103, M106), negative consumption noise (M104), and extreme voltage calibration failure (2300V in M105). We applied equal-frequency binning (depth 4) and bin smoothing to remove granular volatility while preserving underlying trends.

6. Data Integration, Transformation and Correlation

Resolved schema conflicts across university databases (entity resolution, naming conventions, unit conversions, date formats). Computed the Pearson correlation coefficient between student attendance and exam marks (r = 0.9851, indicating a very strong positive linear relationship). Conducted a Chi-Square test of independence between Hostel Status and Scholarship (χ^2 = 33.75 > 3.84, rejecting independence). Demonstrated Min-Max, Z-Score, and decimal scaling normalization.

7. Data Reduction and Discretization

Evaluated industrial sensor reduction strategies (cube aggregation, attribute subset selection, PCA, numerosity reduction). Compared sampling techniques, concluding that stratified sampling is indispensable when rare machine failures constitute only 0.5% of rows. Performed equal-width/frequency discretization and entropy-based split point selection.

8. Day-4 Design Challenge – Loan-Default Pipeline Redesign

Diagnosis of Pipeline Flaws: Our team identified five severe flaws in the baseline pipeline: (1) Global mean income imputation distorts variance; (2) Dropping occupation discards valuable signal; (3) Equal-width binning on skewed income creates empty bins; (4) **Severe data leakage** caused by performing global min-max scaling before the train-test split; (5) Training on an imbalanced dataset (3% defaults) causing model collapse.

Redesigned Pipeline: Stratified train-test split -> Median/KNN imputation on training folds only -> Log transformation -> Z-score scaling on training folds only -> SMOTE oversampling on training data only. Validated via 10-fold cross-validation, achieving 0.84 ROC-AUC and 78% defaulter recall.